

Preservation Bingo — Volunteer Call Sheet

Call a term and read the clue that fits the player age. Check it off as you go.

FOOD SAFETY

☐ **pressure canner** *(equipment)*

Kids: A special big pot that gets very hot to keep food safe when you store it.

Tweens: A pot that uses steam pressure to heat food higher than boiling to kill germs.

Teens: A sealed pot that reaches 240°F using steam pressure to destroy bacteria in low-acid foods.

Adult: A sealed kettle with a lockable lid and accurate pressure gauge, used to heat low-acid foods above boiling temperature (240°F) to destroy *Clostridium botulinum* and other bacteria.

Source: SP 50-1000 Turkey basics

☐ **botulism** *(safety)*

Kids: A very bad sickness you can get from some canned foods if not made right.

Tweens: A serious food poisoning from bacteria that can grow if canned foods aren't heated properly.

Teens: Poisoning caused by *Clostridium botulinum* bacteria that can grow in sealed, low-acid foods not processed at high enough temperatures.

Adult: A life-threatening illness from neurotoxin produced by *Clostridium botulinum* in improperly canned low-acid foods; only pressure canning temperatures (240°F) kill the bacteria and toxin.

Source: SP 50-1000 Turkey basics

☐ **headspace** *(process)*

Kids: The little empty space left on top inside a jar before you close it.

Tweens: The space left between food and jar lid to let food expand when canned or frozen.

Teens: The recommended gap between the food surface and jar rim that allows for expansion and vacuum formation during canning.

Adult: The specified space, usually 1 inch or less, left unfilled in canning jars to allow for food expansion and proper vacuum seal formation during processing.

Source: SP 50-1000 Turkey basics

☐ **boil before eating** *(process)*

Kids: Cook canned food in boiling water before you eat it to keep safe.

Tweens: Boil home canned meats for safety before eating to kill any germs left.

Teens: Boiling home-canned low-acid meats for at least 10 minutes at low altitude destroys residual toxins for safety.

Adult: Boil home-canned meats for 10 minutes at altitudes below 1,000 feet (longer at higher altitudes) as a safety step to destroy any *Clostridium botulinum* toxin possibly present.

Source: SP 50-1000 Turkey basics

☐ **exhaust air** *(process)*

Kids: Let the hot air out of the big pot before cooking cans so food cooks right.

Tweens: Remove air from the pressure canner by letting steam escape for 10 minutes before cooking.

Teens: Venting a pressure canner for 10 minutes before pressurizing removes trapped air for even heating.

Adult: To ensure uniform temperature, air is vented from a pressure canner by opening the petcock and allowing steam to flow for 10 minutes before increasing pressure.

Source: SP 50-1000 Turkey basics

☐ **altitude adjustment** *(safety)*

Kids: Change cooking time for cans if you live somewhere high up, like on a mountain.

Tweens: You must adjust pressure or time when canning at higher places to stay safe.

Teens: Pressure or time must increase in canning at higher altitudes because boiling temperature is lower.

Adult: Increase pressure in weighted-gauge canners above 1,000 ft or dial-gauge canners above 2,000 ft elevation to reach 240°F required for safe processing.

Source: SP 50-1000 Turkey basics

☐ **pH** *(safety)*

Kids: A number that tells if food is sour or not sour, which helps keep it safe.

Tweens: A measure of food's sourness that shows if it's safe to can in water bath or needs pressure canner.

Teens: pH below 4.6 is safe for water bath canning; above requires pressure canning to prevent botulism.

Adult: The numeric measure of acidity; foods with pH ≤ 4.6 are high-acid and safe for boiling water canning, while low-acid foods

(pH > 4.6) require pressure canning to destroy *Clostridium botulinum*.

Source: SP 50-1001 PH values

☐ **pickling salt** (*ingredient*)

Kids: Special salt used to make pickles that helps keep them tasty and safe.

Tweens: Salt without additives made for pickling to keep food safe and clean.

Teens: Pure salt without iodine or anti-caking agents used in pickling to keep food safe and prevent cloudiness.

Adult: Pure granulated sodium chloride salt without iodine or additives used in pickling to ensure clear brine and prevent spoilage.

Source: SP 50-1000 Turkey basics

☐ **sanitizing** (*process*)

Kids: Cleaning things so germs go away.

Tweens: Washing with special soap to kill bad germs that make you sick.

Teens: Using soap or bleach water to kill bacteria and keep your kitchen safe.

Adult: The process of using soap and dilute chlorine bleach or heat to reduce harmful bacteria on kitchen surfaces and equipment to safe levels.

Source: SP 50-1000 Turkey basics

CANNING FRUITS

☐ **sugar syrup** (*ingredient*)

Kids: Sweet water poured on fruit to make it taste yummy in a jar.

Tweens: A mix of sugar and water used to soak fruit before canning.

Teens: A boiling mix of water and sugar poured over fruit in jars to add sweetness and preserve texture.

Adult: A prepared solution of water and measured sugar boiled and poured over fruit in jars, used for sweetness and quality preservation in canning fruits.

Source: PNW 172 Canning vegetables

☐ **hot pack** (*process*)

Kids: Put warm fruit into jars for canning.

Tweens: Fruit is boiled a little before putting it into jars to help preserve it.

Teens: Fruit heated in water, syrup, or juice until hot before packing in jars for canning; reduces floating and shortens processing time.

Adult: A method where fruit is heated (typically boiled briefly) in liquid prior to packing into jars, improving texture, yield, and reducing floating fruit during processing.

Source: PNW 172 Canning vegetables

☐ **raw pack** (*process*)

Kids: Put cold, fresh fruit into jars right away.

Tweens: Put uncooked fruit straight into jars and add hot liquid to keep it safe.

Teens: Fruit packed raw into hot jars then covered with boiling liquid; usually results in firmer texture than hot pack.

Adult: A canning method where raw fruit is packed into hot jars and covered with boiling water, syrup, or juice, usually resulting in firmer texture but more floating fruit than hot pack.

Source: PNW 172 Canning vegetables

☐ **jar lifter** (*equipment*)

Kids: Special tongs to pick up hot jars safely.

Tweens: Tool used to hold and lift hot jars from boiling water.

Teens: Utensil with rubber grips designed to safely remove hot jars from canners without breaking or dropping.

Adult: A specially designed tool with rubber-coated ends used to securely grasp and lift hot canning jars from boiling water or pressure canners during processing.

Source: PNW 172 Canning vegetables

☐ **cooling jars** (*process*)

Kids: Let jars rest and get cooler before touching them.

Tweens: After canning, jars must cool on a towel or rack without moving for a while.

Teens: Jars are placed on a towel or rack undisturbed for 12-24 hours to cool and form a vacuum seal after processing.

Adult: The step where jars are removed from the canner and placed spaced apart on a rack or towel to cool undisturbed for 12 to 24 hours, allowing sealing and preventing spoilage.

Source: PNW 172 Canning vegetables

☐ **acidification** *(safety)*

Kids: Adding something sour to keep fruit safe in jars.

Tweens: Put lemon juice or vinegar in fruit jars to make them safe by making the jar more sour.

Teens: Adding lemon juice or citric acid to fruits to lower pH and prevent harmful bacteria growth during canning.

Adult: The process of adding acid substances such as bottled lemon juice or citric acid to fruits to ensure the pH is low enough to inhibit pathogenic bacteria, making the product safe for boiling water canning.

Source: PNW 172 Canning vegetables

☐ **processing time** *(process)*

Kids: How long we keep jars hot so fruits stay good.

Tweens: The minutes jars stay in boiling water or canner to keep fruit safe.

Teens: The scientifically tested minutes jars need heat treatment in a canner to kill microbes and ensure safe preservation.

Adult: A research-based duration for heat processing jars in boiling water or pressure canners to destroy spoilage organisms and pathogens, varying by fruit type, jar size, and altitude.

Source: PNW 172 Canning vegetables

☐ **removing air bubbles** *(process)*

Kids: Poke the jars to pop tiny air bubbles out.

Tweens: Use a tool to take air bubbles out of jars before sealing so fruit fits right.

Teens: Running a plastic tool inside filled jars to release trapped air bubbles that can affect sealing and preservation.

Adult: The step of gently moving a non-metallic utensil inside filled jars to release trapped air bubbles to prevent spoilage and ensure correct headspace and sealing.

Source: PNW 172 Canning vegetables

☐ **syrup** *(ingredient)*

Kids: A sweet juice that makes fruit yummy in jars.

Tweens: A sugary liquid poured on fruit to keep it sweet and tasty.

Teens: A sugar and water mixture poured over fruit before canning to add sweetness and preserve texture.

Adult: A sugar syrup is a heated solution of sugar and water used in fruit canning to enhance flavor, texture, and preservation, with sweetness levels from very light to heavy.

Source: PNW 199 Canning fruits

CANNING TOMATOES & SALSAS

☐ **food mill** *(equipment)*

Kids: A tool to mash tomatoes like a big strainer.

Tweens: You use it to press cooked tomatoes to get a smooth sauce without seeds or skin.

Teens: A kitchen device that purees and sieves cooked tomatoes, removing seeds and skins for smooth sauces.

Adult: A food mill is a hand-cranked utensil used to press cooked tomatoes through a fine sieve, removing skins and seeds, producing a smooth texture before canning.

Source: PNW 172 Canning vegetables

☐ **vinegar (5% acid)** *(ingredient)*

Kids: A sour liquid that helps keep salsa safe.

Tweens: We add a special sour juice called vinegar to stop bad germs growing in salsa.

Teens: Vinegar with 5% acidity is added to salsa to lower pH and ensure safety by preventing harmful bacteria.

Adult: Vinegar at 5% acidity is used in salsa canning to acidify the product sufficiently to inhibit growth of *Clostridium botulinum* and other microbes, enabling boiling water canning.

Source: PNW 395 Salsa recipes for canning

☐ **simmer** *(process)*

Kids: Cooking food gently on the stove.

Tweens: Cooking the tomatoes or salsa slowly with small bubbles so it doesn't burn.

Teens: Simmering is heating a sauce just below boiling to blend flavors and soften ingredients without harsh boil.

Adult: Simmering involves heating the tomato or salsa mixture below boiling point (about 185-205°F) to meld flavors, reduce volume, and soften components before packing.

Source: PNW 395 Salsa recipes for canning

☐ **chile peppers** *(ingredient)*

Kids: Spicy peppers that make salsa hot.

Tweens: Chiles add heat and flavor to your salsa recipes.

Teens: Chile peppers, either mild or hot, are chopped and added to salsas for flavor and heat; they require careful handling.

Adult: Chopped chile peppers, both mild and hot varieties, are used in tomato salsas to add heat and flavor; gloves should be worn when handling to prevent irritation.

Source: PNW 300 Canning tomatoes and tomato products

☐ **cold water bath** (process)

Kids: Put jars in cool water to make the lid stick.

Tweens: After canning, cool jars in water so lids seal tight and food stays safe.

Teens: Cooling jars quickly in a cold water bath helps set the vacuum seal and prevents spoilage.

Adult: After processing, jars are removed and gradually cooled, often by placing in cold water or on a cooling rack, to ensure proper vacuum seal formation and prevent thermal shock.

Source: PNW 300 Canning tomatoes and tomato products

☐ **tomato paste** (ingredient)

Kids: Thick tomato sauce that makes salsa thicker.

Tweens: Tomato paste is cooked-down tomatoes that add thick texture to sauces and salsas.

Teens: Tomato paste is concentrated cooked tomatoes used as a base for thicker salsas and sauces.

Adult: Tomato paste, a cooked and concentrated tomato product, is used to thicken salsas and sauces, enhancing consistency and flavor intensity.

Source: PNW 395 Salsa recipes for canning

☐ **lemon juice** (ingredient)

Kids: A sour juice that makes tomatoes safe to save in jars.

Tweens: Adds sour juice to tomatoes to keep them safe when canned.

Teens: Used to acidify tomatoes and salsas to prevent spoilage during canning.

Adult: Bottled lemon or lime juice added to tomatoes or salsa to reach a safe acidity level for boiling water canning, typically about 1 1/2 cups per recipe.

Source: PNW 395 Salsa recipes for canning

☐ **boiling water canner** (equipment)

Kids: Big pot with lid that boils jars to keep food safe.

Tweens: A special pot that boils jars to kill germs in canned tomatoes and salsas.

Teens: A large pot with a lid used to process high-acid foods like tomatoes and salsas by boiling to destroy spoilage organisms.

Adult: A canning device that processes acidified foods (like tomatoes and salsas) using boiling water for specified times adjusted by altitude to ensure safe preservation.

Source: PNW 395 Salsa recipes for canning

☐ **citric acid** (ingredient)

Kids: A special powder that makes tomatoes safe and yummy when you can them.

Tweens: A powder that helps stop germs and keeps canned tomatoes safe to eat.

Teens: Citric acid is added to canned tomatoes to increase acidity and prevent harmful bacteria growth.

Adult: Citric acid USP, typically 1/4 teaspoon per pint jar, is added to tomatoes during canning to ensure the product's pH is sufficiently low (below 4.6) for safe boiling-water canning and to prevent Clostridium botulinum growth.

Source: PNW 300 Canning tomatoes and tomato products

LOW ACID FOODS

☐ **pressure canner weight** (equipment)

Kids: The heavy part on a pot that helps cook food safely with steam.

Tweens: This weight on the canning pot lets steam out and keeps the right cooking pressure.

Teens: A weighted gauge on a pressure canner controls pressure by releasing steam, usually 10 or 15 pounds depending on altitude.

Adult: A weighted gauge pressure canner uses a specific weight (10 or 15 lb) on the vent pipe to regulate steam pressure for safe low-acid food canning and varies by elevation.

Source: PNW 421 Use care operation of your pressure canner

☐ **exhausting air** (process)

Kids: Letting out all the air before cooking to make food safe.

Tweens: Releasing air from the canner so steam can heat food evenly.

Teens: Venting the pressure canner by letting steam escape for 10 minutes removes trapped air that could prevent proper heating.

Adult: Exhausting air involves venting all types of pressure canners by allowing steam to escape steadily for 10 minutes before pressure builds to ensure uniform temperature and safe processing.

Source: PNW 172 Canning vegetables

☐ **safe processing time** (safety)

Kids: Cooking food long enough to kill germs so you don't get sick.

Tweens: Following the right cooking time so all dangerous things in food are gone.

Teens: Processing low-acid foods in a pressure canner requires exact times to kill bacteria like *Clostridium botulinum*.

Adult: Safe processing time is the researched duration at specific pressures and temperatures necessary to destroy pathogens and spoilage organisms in low-acid foods during pressure canning.

Source: PNW 361 Canning meat poultry game

☐ **clostridium botulinum** (safety)

Kids: A tiny bug that can make people very sick if food isn't cooked right.

Tweens: A bacteria that grows on low-acid foods and makes dangerous poison if food is not canned safely.

Teens: A spore-forming bacteria causing botulism, which can survive in improperly canned low-acid foods, producing deadly toxin.

Adult: *Clostridium botulinum* is an anaerobic, spore-forming bacterium producing a potent neurotoxin; it can survive in sealed low-acid foods unless processed above 240°F via pressure canning.

Source: PNW 361 Canning meat poultry game

☐ **jar headspace** (process)

Kids: The empty space left at the top of a jar to keep food safe.

Tweens: Leaving a bit of space at the top of jars so food cooks well and seals right.

Teens: Headspace prevents food from touching the lid by leaving a specific empty space; it's critical for vacuum sealing and safe heating.

Adult: Headspace is the space between the packed food and the jar lid, usually 1 inch for low-acid foods, to allow for expansion and vacuum formation during processing.

Source: PNW 172 Canning vegetables

☐ **pressure gauge** (equipment)

Kids: The little dial telling how strong the canning pot is cooking food.

Tweens: A dial on a canner that shows when the pressure is just right for cooking.

Teens: Dial-gauge pressure canners use a gauge to monitor pressure; maintaining correct psi is key for safe low-acid food processing.

Adult: Pressure gauge is an instrument on dial-gauge pressure canners that indicates internal pressure; requires annual calibration to ensure food safety.

Source: PNW 361 Canning meat poultry game

☐ **cool down period** (process)

Kids: Waiting time after cooking before opening the jar, like a rest.

Tweens: Letting the canner cool slowly so food finishes cooking safely inside the jar.

Teens: Allowing the pressure canner to cool naturally helps continue killing bacteria; quick cooling is unsafe.

Adult: Cool down period is the time after processing during which the canner cools naturally; continued microbial destruction occurs and sudden depressurizing is avoided.

Source: PNW 361 Canning meat poultry game

☐ **oven canning** (safety)

Kids: Cooking food in an oven to store it, which is not safe.

Tweens: Using an oven to can food is risky and not recommended for safety.

Teens: Oven canning does not reach safe temperatures and can allow botulism spores to survive.

Adult: Oven canning is unsafe as it does not reach the required internal temperatures (>240°F) to destroy *Clostridium botulinum* spores in low-acid foods.

Source: PNW 361 Canning meat poultry game

☐ **pressure canning** (process)

Kids: Cooking low acid foods in a special pot to keep them safe.

Tweens: Using a special cooker that gets really hot to kill germs in low acid foods.

Teens: Heating low acid foods in a pressure canner to reach 240°F, killing bacteria like *Clostridium botulinum*.

Adult: Processing low acid foods (pH above 4.6) in a pressure canner at 240°F to destroy *Clostridium botulinum* spores that boiling water canners cannot eliminate, ensuring food safety.

Source: PNW 361 Canning meat poultry game

PICKLING

☐ **pickling lime** (ingredient)

Kids: A special powder that helps make pickles crunchy.

Tweens: We soak cucumbers in this powder water to make pickles extra firm.

Teens: Pickling lime is used to soak cucumber slices before pickling to improve crispness by firming the texture.

Adult: Pickling lime (calcium hydroxide) is used in a lime-water solution soak to firm cucumbers by increasing calcium content before pickling, requiring careful rinsing after.

Source: PNW 355 Pickling vegetables

☐ **dill heads** *(ingredient)*

Kids: Green fluffy plants that give your pickles a yummy taste.

Tweens: We add dill heads to jars to make pickles smell and taste good.

Teens: Dill heads are aromatic herbs placed in jars to flavor pickled beans or cucumbers.

Adult: Dill heads or sprigs are fresh aromatic herbs used in pickling to impart characteristic flavor and aroma to pickled vegetables, often used in hot-pack processes.

Source: PNW 355 Pickling vegetables

☐ **pickling spice** *(ingredient)*

Kids: A colorful mix of seeds and spices that make pickles taste yummy.

Tweens: Pickling spice is a blend of spices you add to jars for extra flavor in pickles.

Teens: Pickling spice is a mixture of various whole spices like mustard seed, cinnamon, and cloves used to flavor pickled products.

Adult: Pickling spice is a blend of whole spices-often mustard seed, cinnamon sticks, cloves, and others-used to impart characteristic flavors to pickled vegetables.

Source: PNW 355 Pickling vegetables

☐ **lime-water soak** *(process)*

Kids: A special bath that makes pickles extra crisp and crunchy.

Tweens: Soaking vegetables in lime water helps make them firm for pickling.

Teens: A soak in lime-water solution firms cucumbers by strengthening their texture before pickling.

Adult: Soaking cucumbers in a solution of pickling lime and water for 12-24 hours strengthens cell walls to produce firmer pickles; this requires subsequent multiple rinsings to remove excess lime.

Source: PNW 355 Pickling vegetables

☐ **blanching** *(process)*

Kids: Quick splash of hot water to help veggies stay bright and safe.

Tweens: Heating vegetables briefly in hot water keeps them crisp and kills some germs.

Teens: Blanching is briefly boiling produce before pickling to soften it and reduce spoilage microbes.

Adult: Blanching involves briefly immersing vegetables in boiling water or steam, then rapidly cooling them to inactivate enzymes and reduce microbial load prior to pickling.

Source: PNW 355 Pickling vegetables

☐ **dill pickles** *(result)*

Kids: Pickles flavored with spicy green plants called dill.

Tweens: Pickles made with dill have a special tangy taste kids love.

Teens: Dill pickles are cucumbers pickled with dill weed or heads to give a specific flavor.

Adult: Dill pickles are cucumbers preserved in vinegar with dill herb-stalks, heads, or seeds-providing distinctive flavor, commonly processed by boiling water bath canning.

Source: PNW 355 Pickling vegetables

☐ **pickle fermenting crock** *(equipment)*

Kids: A big pot where you let pickles change and get yummy on their own.

Tweens: A special jar or pot where vegetables slowly change into pickles with good germs.

Teens: A fermenting crock is a container used for lacto-fermentation of pickles to allow natural acid development without spoilage.

Adult: A fermenting crock is a non-reactive, often ceramic or enamelware container designed to provide optimum conditions for vegetable fermentation by lacto-bacilli during pickle making.

Source: PNW 355 Pickling vegetables

☐ **brine** *(ingredient)*

Kids: A salty water that helps keep pickles tasty and fresh.

Tweens: A salty and sour water that helps vegetables stay crunchy and safe to eat.

Teens: A solution of salt and vinegar (sometimes water) that preserves vegetables by creating an acidic environment to prevent spoilage.

Adult: A pickling solution typically made of vinegar (5% acidity), salt, and sometimes water and spices, used to preserve vegetables by acidification preventing microbial growth.

Source: PNW 355 Pickling vegetables

☐ **mustard seed** *(ingredient)*

Kids: A tiny yellow seed that gives pickles a spicy taste.

Tweens: Small seeds that add a special spicy flavor to pickles.

Teens: Mustard seeds are used in pickling to add flavor and a mild spice note.

Adult: Mustard seed is a common pickling spice ingredient contributing pungent, spicy flavor to pickled products, found in many vegetable pickle recipes.

Source: PNW 355 Pickling vegetables

JAMS & JELLIES

☐ **pectin** *(ingredient)*

Kids: A special powder from fruit that helps jams and jellies become thick and jelly-like.

Tweens: Pectin is a natural part of fruit that helps your jelly and jam set and not be runny.

Teens: Pectin is a natural substance in fruit that gels jams and jellies; commercial pectin is often added to help jams set.

Adult: Pectin, a natural polysaccharide in fruit, causes gel formation in jams and jellies. Commercial pectin (powdered or liquid) is used to ensure proper gel formation and must be used as per manufacturer's instructions and processing times.

Source: SP 50-632 Making berry syrups at home

☐ **gel stage** *(process)*

Kids: The magic moment when jam gets thick and jelly jiggles just right.

Tweens: Gel stage is when you cook the jelly or jam until it becomes thick and holds its shape.

Teens: The gel stage occurs when jam or jelly reaches a temperature or consistency where it thickens and sets, usually tested by the sheet or wrinkle test.

Adult: Gel stage is reached when a heated jam or jelly mixture achieves the proper temperature (about 220°F or 104°C) or demonstrates gel formation evidenced by the sheet test, indicating sugar-pectin-acid interactions have formed a stable gel.

Source: SP 50-632 Making berry syrups at home

☐ **foam** *(result)*

Kids: Bubbles that form on top of jam juice when it cooks.

Tweens: Foam is the bubbly stuff that appears on jam or jelly as it cooks and must be removed.

Teens: Foam is protein or air bubbles forming on the surface of cooking jam or jelly; removing it improves texture and clarity.

Adult: During jam or jelly boiling, foam composed of proteins and trapped air forms on the surface; skimming off foam just before jarring improves the final product's appearance and reduces microbial contamination risk.

Source: SP 50-632 Making berry syrups at home

☐ **fruit spread** *(result)*

Kids: Yummy sweet fruit you can spread on toast that isn't always hard like jelly.

Tweens: Fruit spreads are sweet fruit mixtures like jam or jelly, sometimes thicker and smoother to spread on bread.

Teens: Fruit spreads are sweetened fruit products including jam, jelly, marmalade, and preserves that vary in texture and fruit content.

Adult: Fruit spreads encompass jams, jellies, preserves, marmalades, and butters, made from fruit, sugar, pectin, and acid, differing in texture and fruit content according to preparation method.

Source: SP 50-632 Making berry syrups at home

☐ **low methoxyl pectin** *(ingredient)*

Kids: A special kind of pectin that helps make jam with less sugar.

Tweens: Low methoxyl pectin helps jams set well even when you use less sugar or no sugar.

Teens: Low methoxyl pectin is a type of pectin that forms gels in jams with less sugar and often requires calcium to set.

Adult: Low methoxyl pectin is a modified pectin that gels in the presence of calcium ions and lower sugar concentrations, allowing the production of low or no sugar jams and jellies with a different texture and set time.

Source: SP 50-632 Making berry syrups at home

☐ **freezer jam** *(process)*

Kids: Jam you make and keep in the freezer instead of canning it.

Tweens: Freezer jams are jams you make without cooking and keep in the freezer or fridge to stay fresh.

Teens: Freezer jams use uncooked fruit and pectin, stored in the freezer or refrigerator without boiling water canning for a fresher taste but shorter shelf life.

Adult: Freezer jams are uncooked jams made with powdered pectin and fresh or frozen fruit sugar mixture, packaged and stored in freezer or refrigerator; they are not heat processed and have limited shelf life.

Source: SP 50-632 Making berry syrups at home

☐ **acid** *(ingredient)*

Kids: A sour food like lemon juice that helps jam set and taste good.

Tweens: Acid like lemon juice helps jams and jellies set by helping pectin gel and makes the flavor bright.

Teens: An acid, such as lemon juice, is added to jams and jellies to balance pH for gel formation and to enhance flavor and preservation.

Adult: Acid, often from lemon juice or fruit acids, lowers pH in jams and jellies to optimal levels for pectin gelation and improves microbial safety and palatability.

Source: SP 50-632 Making berry syrups at home

☐ **skim foam** *(process)*

Kids: Taking away the bubbles on top of jam so it looks nice.

Tweens: Skim foam means using a spoon to take off bubbles that form on top when making jam or jelly.

Teens: Skimming foam involves removing the protein-rich bubbles that form on the surface as jams and jellies cook to improve quality and appearance.

Adult: The process of skimming foam during jam or jelly making removes surface proteins and impurities, improving clarity, texture, and reducing microbial contamination.

Source: SP 50-632 Making berry syrups at home

☐ **skimming foam** *(process)*

Kids: Taking off the bubbles on top when cooking jam.

Tweens: You remove the frothy bubbles on jam's surface so it looks clear and neat.

Teens: Skimming foam means quickly removing the bubbly layer that forms on the surface during cooking jams or jellies to improve texture and appearance.

Adult: Skimming foam is the process of quickly removing the protein and air bubbles that rise to the surface while cooking jams or jellies, preventing defects in gel clarity and texture before pouring into jars.

Source: SP 50-632 Making berry syrups at home

FREEZING & COLD STORAGE

☐ **freezer burn** *(result)*

Kids: Dry spots on food from the freezer.

Tweens: When freezer air makes food dry and fuzzy.

Teens: Damage from air causing dehydration and spots on frozen food.

Adult: Surface dehydration and oxidation damage to frozen food caused by air exposure, reducing quality.

Source: PNW 586 Home freezing seafood

☐ **glazing** *(process)*

Kids: Coating food with a shiny ice layer.

Tweens: Dipping frozen fish in water to coat with ice.

Teens: Protective ice coating applied to frozen seafood by dipping in cold water.

Adult: Dipping frozen seafood in cool water to form an ice coating that protects against freezer burn during storage.

Source: PNW 586 Home freezing seafood

☐ **freezer storage** *(process)*

Kids: Keeping food very cold to save it.

Tweens: Putting food in the freezer so it stays good for a long time.

Teens: Storing food at temperatures below freezing to preserve quality and safety by slowing bacteria growth.

Adult: Maintaining food at or below 0 °F (-18 °C) to preserve quality and safety by halting bacterial activity and enzymatic changes.

Source: SP 50-711 Frozen food storage guide

☐ **freezer burn protection** *(process)*

Kids: Keeping air away from frozen food so it stays nice.

Tweens: Wrapping food tight so freezer air won't dry it out.

Teens: Excluding air from packaging slows freezing, reduces ice crystals and freezer burn.

Adult: Removing air from packaging to minimize oxidation and large ice crystals that cause dehydration and texture damage.

Source: PNW 586 Home freezing seafood

☐ **freezer temperature** *(safety)*

Kids: Keep the freezer very cold to keep food safe.

Tweens: Freezers need to stay very cold to stop germs from growing.

Teens: Storage below 0°F (-18°C) is required to maintain food safety and quality during freezing.

Adult: Maintaining freezer temperature at or below 0°F (-18°C) ensures bacterial growth stops and food remains safe during

storage.

Source: PNW 250 You can prevent food borne illness

☐ **airtight container** (equipment)

Kids: A special box that keeps air out to keep food fresh.

Tweens: A container that locks air out so food stays good in the freezer.

Teens: Containers that seal tightly to prevent air exposure and moisture loss in frozen foods.

Adult: Containers with tight-fitting lids designed to exclude air, preventing oxidation and freezer burn during frozen storage.

Source: PNW 586 Home freezing seafood

☐ **freezing rate** (process)

Kids: How fast food gets really cold and frozen.

Tweens: Fast freezing makes smaller ice crystals, protecting food quality.

Teens: Rapid freezing forms small ice crystals that cause less damage to food texture than slow freezing.

Adult: The speed at which food temperature drops; rapid freezing decreases ice crystal size, preserving texture and quality.

Source: PNW 586 Home freezing seafood

☐ **frost-free freezer** (equipment)

Kids: A freezer that stops ice from building up inside.

Tweens: A freezer that automatically stops frost and keeps food dry.

Teens: Freezer with automatic defrost cycles preventing frost build-up and moisture damage.

Adult: A freezer that periodically melts frost formation automatically to prevent ice build-up and maintain efficient operation.

Source: OSU Extension MFP

☐ **fast freezing** (process)

Kids: Freezing food quickly so it stays yummy and fresh.

Tweens: Freezing food fast to keep it tasting good and stop ice crystals from getting big.

Teens: A quick freezing method that forms smaller ice crystals to reduce cell damage and keep food quality better.

Adult: Fast freezing is freezing foods rapidly to form small ice crystals, minimizing cell destruction and preserving quality during storage; slow freezing forms larger crystals that damage cells and reduce quality.

Source: PNW 586 Home freezing seafood

DEHYDRATING & SMOKING

☐ **dehydrator** (equipment)

Kids: A machine that dries your fruits and veggies.

Tweens: A box that blows warm air to dry foods like fruit and vegetables.

Teens: An appliance that removes moisture by circulating warm air over food to dry it safely.

Adult: Equipment that distributes warm air at controlled temperatures (140-150°F initially) to remove moisture from foods without losing texture or flavor.

Source: PNW 397 Drying fruits and vegetables

☐ **marinade** (ingredient)

Kids: A yummy sauce for meat before drying.

Tweens: A flavored liquid that meats soak in before drying for taste.

Teens: A seasoned liquid mix used to soak meat to add flavor and sometimes tenderize before drying or smoking.

Adult: A seasoning liquid, often containing salt, sugar, and spices, used to flavor meat prior to drying; must be boiled with meat strips to maintain safety.

Source: PNW 632 Making jerky at home safely

☐ **hot pickle cure** (process)

Kids: A special salty bath for meat before drying.

Tweens: A salt and spice cure for meat done before drying to keep it safe and tasty.

Teens: A two-step curing method involving salting meat, then simmering in brine before drying for safety and flavor.

Adult: A two-step precooking involving 24-hour refrigeration with salt-sugar-pepper cure, followed by simmering meat strips in hot brine for 1.5-2 minutes before drying.

Source: PNW 632 Making jerky at home safely

☐ **smokehouse** (equipment)

Kids: A house where meat gets smoky flavor.

Tweens: A special shelter to hang meat for smoking.

Teens: A controlled structure or chamber used to expose foods to smoke for preservation and flavoring.

Adult: An insulated enclosure designed to hold meats/fish at regulated temperatures during prolonged exposure to smoke for curing

and drying.

Source: PNW 238 Smoking fish at home

☐ **precooking** *(process)*

Kids: Cooking meat a little before drying it.

Tweens: Cooking meat partway to make it safe before drying.

Teens: Heating meat to a safe internal temperature before drying to reduce bacteria risk.

Adult: Heating meat strips to 160°F (165°F for poultry) by boiling marinade or baking before drying to ensure microbial safety.

Source: PNW 632 Making jerky at home safely

☐ **drying time** *(process)*

Kids: How long it takes for food to get dry.

Tweens: The number of hours food needs to dry safely.

Teens: The duration food spends in the dehydrator or smokehouse until moisture is removed adequately.

Adult: The time necessary at controlled temperatures with good air circulation to reduce moisture content to safe storage levels without spoilage.

Source: PNW 397 Drying fruits and vegetables

☐ **jerky** *(result)*

Kids: Dry tasty strips of meat.

Tweens: Thin strips of meat dried for a chewy snack.

Teens: Lean meat that's been seasoned, precooked, and dried to preserve it.

Adult: Lean, thinly sliced meat seasoned, precooked to 160°F, dried until safe, moisture low enough to inhibit microbial growth.

Source: PNW 632 Making jerky at home safely

☐ **airflow** *(safety)*

Kids: Air that moves to dry food faster.

Tweens: Moving air in a dryer helps dry food quicker and safer.

Teens: Circulating air removes moisture, speeds drying, and helps prevent spoilage.

Adult: Adequate air circulation during drying is crucial to remove moisture evenly and reduce drying time, lowering spoilage risks.

Source: PNW 397 Drying fruits and vegetables

☐ **pellicle** *(process)*

Kids: A sticky skin that helps smoke stick to fish.

Tweens: It's a shiny layer on fish that helps keep it safe and tastier when smoked.

Teens: A pellicle is a sticky, dryish coating on fish that forms after brining, helping smoke stick and preventing spoilage during smoking.

Adult: A pellicle is a shiny, slightly tacky skin that forms on the surface of fish after air-drying post-brining, which facilitates even smoke deposition during smoking and helps prevent surface spoilage.

Source: PNW 238 Smoking fish at home